

## Polar Unit Test (Practice D)

## Calculator

You have the entire period.

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test. \_\_\_\_\_

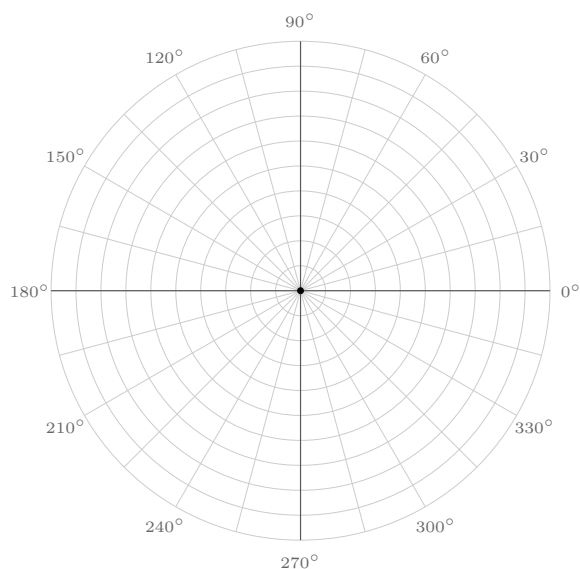
SHOW ALL WORK FOR FULL CREDIT.

Signature \_\_\_\_\_

Use your calculator to sketch a graph of each polar equation. Identify the type of polar graph it is. Show your set up (what you are putting into the calculator). (3 points each)

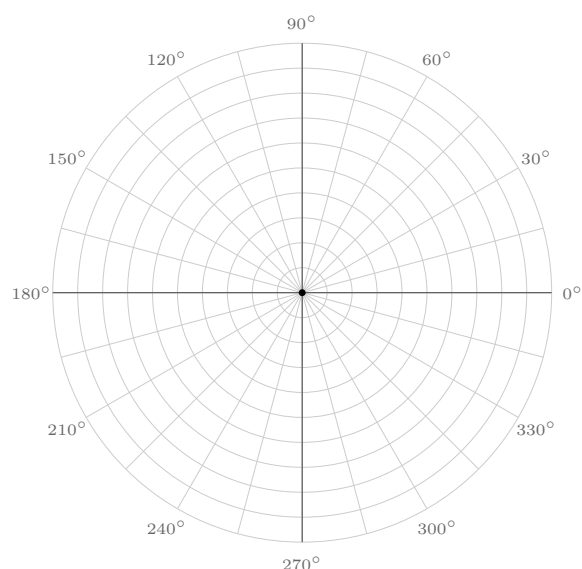
**Word bank:** circle    cardioid    limaçon    rose curve    lemniscate

1.  $r = 3 - 3 \sin \theta$



Type: \_\_\_\_\_

2.  $r^2 = 16 \cos 2\theta$



Type: \_\_\_\_\_

## Polar Unit Test (Practice D)

## Non-Calculator

SHOW ALL WORK FOR FULL CREDIT.

|                                                                                                                                                      |                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Simplify. (3 points)</b><br>3. $\frac{\sqrt{14}}{\sqrt{-2} \cdot \sqrt{-21}}$                                                                     | <b>Simplify. (2 points)</b><br>4. $\frac{-12 + \sqrt{-20}}{8}$                                                                   |
| <b>Perform the indicated operation. Write the answer in rectangular form. (2 points)</b><br>5. $i^{210}$                                             | <b>Perform the indicated operation. Write the answer in rectangular form. (2 points)</b><br>6. $(6 - 4i) - (-2 - 7i) - (9 + 2i)$ |
| <b>Perform the indicated operation. Write the answer in rectangular form. (3 points)</b><br>7. $\frac{\sqrt{27} - i\sqrt{27}}{\sqrt{3} + i\sqrt{3}}$ | <b>Perform the indicated operation. Write the answer in rectangular form. (3 points)</b><br>8. $-2i(1 + 3i)(2 - i)$              |
| <b>Solve each equation over the set of complex numbers. (3 points)</b><br>9. $x^2 + 6x = -10$                                                        |                                                                                                                                  |

**Convert to trig form. (3 points)**

10.  $3 - 3i$

**Convert to rectangular form. (3 points)**

11.  $10 \operatorname{cis} \frac{5\pi}{3}$

**Using  $z^2 - 1, (z^2 - 1)^2 - 1, [(z^2 - 1)^2 - 1]^2 - 1, \dots$  is the following in the Julia set? Show ALL work. (3 points)**

12.  $z = 1$

YES / NO

**Perform the indicated operation. Write each answer in rectangular form. (3 points each)**

13.  $\frac{8(\cos 40^\circ + i \sin 40^\circ)}{4(\cos 250^\circ + i \sin 250^\circ)}$

14.  $\left[4 \operatorname{cis} \frac{5\pi}{3}\right] \left[3 \operatorname{cis} \frac{5\pi}{6}\right]$

Using  $z, z^2 + z, (z^2 + z)^2 + z, \dots$  is the following number in the Mandelbrot set? Show ALL work. (3 points)

15.  $z = -2$

YES / NO

Find each power. Write your answer in rectangular form. (3 pts)

16.  $(-\sqrt{3} + i)^6$

Find all complex number solutions of each equation. Leave answers in trigonometric form. (4 pts)

17.  $x^4 + 16 = 0$

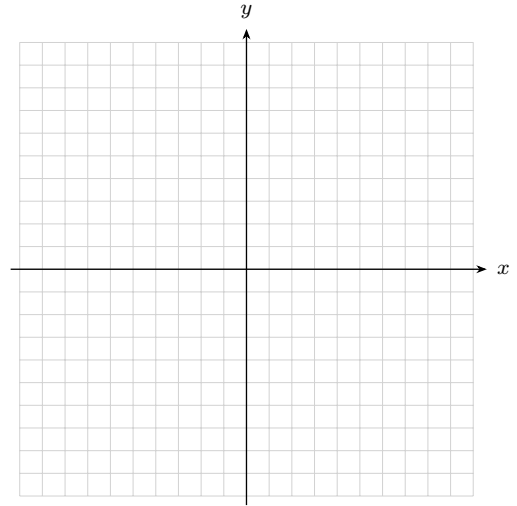
|                                                                                                                                                                  |                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Convert the polar coordinates to rectangular coordinates. Write your answer in exact form. (3 points)</b></p> <p>18. <math>(-6, \frac{5\pi}{6})</math></p> | <p><b>Convert the rectangular coordinates to polar coordinates with <math>r &gt; 0</math> and <math>0^\circ \leq \theta &lt; 360^\circ</math>. (3 points)</b></p> <p>19. <math>(-4, -4)</math></p> |
| <p><b>Convert the rectangular equation to a polar equation. Solve for <math>r</math>. (3 points)</b></p> <p>20. <math>3x - 4y = 12</math></p>                    | <p><b>Find the polar equation of the line that passes through the two points. Simplify your answer. (3 points)</b></p> <p>21. <math>(5, 0^\circ)</math> and <math>(-5, 90^\circ)</math></p>        |

**Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)**

22.  $r = -6 \cos \theta$

Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

23.  $r = \frac{8}{\cos \theta - 2 \sin \theta}$



Find the polar coordinates of the points of intersection for the given interval of  $\theta$ . (4 points)

24.  $r = 3 + 2 \sin \theta$ ,  $r = 3 + 2 \cos \theta$   
 $0 \leq \theta < 2\pi$

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (2 points)

25.  $r = 4 + 3 \sin \theta$

Type: \_\_\_\_\_

**ANSWER KEY – Calculator**

*Left: the work, one step per line, the way it goes on paper. Right: why that line happens.*

*Pace check: about 50 minutes for a strong pass. Calculator part about 5, #3–12 about 15, #13–17 about 13, #18–25 about 15. That leaves about 30 minutes of the period to check your work.*

Use your calculator to sketch a graph of the polar equation. Identify the type of polar graph it is. (3 points)

**1.**  $r = 3 - 3 \sin \theta$

WORK

Polar mode:  $r1(\theta) = 3 - 3 \sin \theta$

$90^\circ: r = 0; 270^\circ: r = 6; 0^\circ, 180^\circ: r = 3$

$$\frac{a}{b} = \frac{3}{3} = 1$$

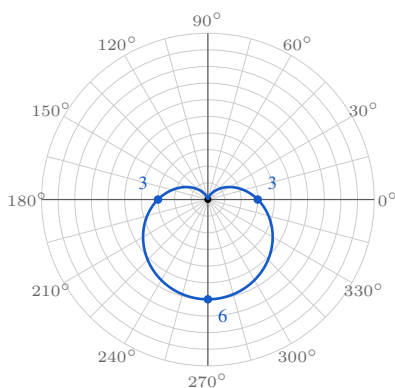
cardioid

WHY

Degree mode needs  $\theta$  from 0 to 360 (radians: 0 to  $2\pi$ ).

Touches the pole at  $90^\circ$ .

Ratio 1.



Use your calculator to sketch a graph of the polar equation. Identify the type of polar graph it is. (3 points)

**2.**  $r^2 = 16 \cos 2\theta$

WORK

$r1(\theta) = \sqrt{16 \cos(2\theta)}$

$\theta = 0^\circ: r = 4; 180^\circ: r = 4$

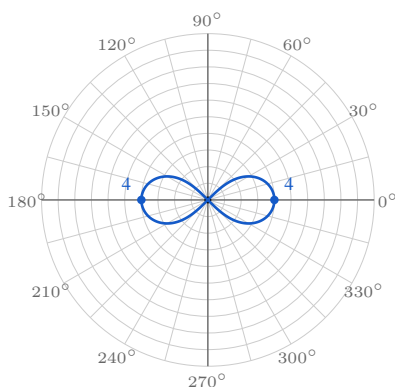
$\cos 2\theta < 0$  for  $45^\circ < \theta < 135^\circ$ : no graph

lemniscate

WHY

Solve for  $r$  first; one root is enough.

Loops along the  $x$ -axis.



## ANSWER KEY – Non-Calculator

Left: the work, one step per line, the way it goes on paper. Right: why that line happens.

Simplify. (3 points)

3.  $\frac{\sqrt{14}}{\sqrt{-2} \cdot \sqrt{-21}}$

WORK

$$\frac{\sqrt{14}}{i^2 \sqrt{42}} = -\sqrt{\frac{14}{42}} = -\frac{1}{\sqrt{3}}$$

$$= \boxed{-\frac{\sqrt{3}}{3}}$$

Trap: Writing  $\sqrt{-2} \cdot \sqrt{-21} = \sqrt{42}$  flips the sign.

WHY

Pull out  $i$  first; reduce inside one radical.

Rationalize.

Simplify. (2 points)

4.  $\frac{-12 + \sqrt{-20}}{8}$

WORK

$$\sqrt{-20} = 2i\sqrt{5}$$

$$\frac{-12 + 2i\sqrt{5}}{8} = \boxed{-\frac{3}{2} + \frac{\sqrt{5}}{4}i}$$

WHY

Reduce each part.

Perform the indicated operation. Write the answer in rectangular form. (2 points)

5.  $i^{210}$

WORK

$$210 \div 4 = 52 \text{ R } 2$$

$$i^2 = \boxed{-1 + 0i}$$

WHY

Perform the indicated operation. Write the answer in rectangular form. (2 points)

6.  $(6 - 4i) - (-2 - 7i) - (9 + 2i)$

WORK

$$6 - 4i + 2 + 7i - 9 - 2i$$

$$\boxed{-1 + i}$$

WHY

Flip both parts of each subtracted group.

real  $-1$ , imag  $1$ .

Perform the indicated operation. Write the answer in rectangular form. (3 points)

7.  $\frac{\sqrt{27} - i\sqrt{27}}{\sqrt{3} + i\sqrt{3}}$

WORK

$$\cdot \frac{\sqrt{3} - i\sqrt{3}}{\sqrt{3} - i\sqrt{3}}$$

top:  $9 - 9i - 9i + 9i^2 = -18i$

bottom:  $3 + 3 = 6$

$$\boxed{0 - 3i}$$

WHY

Conjugate of the bottom.

$$\sqrt{27}\sqrt{3} = 9.$$

Perform the indicated operation. Write the answer in rectangular form. (3 points)

8.  $-2i(1 + 3i)(2 - i)$

WORK

$$(1 + 3i)(2 - i) = 2 - i + 6i - 3i^2 = 5 + 5i$$

$$-2i(5 + 5i) = -10i - 10i^2 = \boxed{10 - 10i}$$

WHY

Solve each equation over the set of complex numbers. (3 points)

9.  $x^2 + 6x = -10$

WORK

$$x^2 + 6x + 10 = 0$$

$$x = \frac{-6 \pm \sqrt{36 - 40}}{2} = \frac{-6 \pm 2i}{2}$$

$$\boxed{x = -3 \pm i}$$

WHY

Convert to trig form. (3 points)

10.  $3 - 3i$

WORK

$$r = \sqrt{9 + 9} = 3\sqrt{2}$$

ref  $45^\circ$ , QIV:  $315^\circ$

$$\boxed{3\sqrt{2}(\cos 315^\circ + i \sin 315^\circ)}$$

WHY

Convert to rectangular form. (3 points)

11.  $10 \operatorname{cis} \frac{5\pi}{3}$

WORK

$$10 \left( \frac{1}{2} - \frac{\sqrt{3}}{2}i \right) = \boxed{5 - 5\sqrt{3}i}$$

WHY

QIV.

Using  $z^2 - 1$ ,  $(z^2 - 1)^2 - 1$ ,  $[(z^2 - 1)^2 - 1]^2 - 1$ , ... is the following in the Julia set? Show ALL work. (3 points)

**12.**  $z = 1$

YES / NO

WORK

$$1^2 - 1 = 0, 0^2 - 1 = -1, (-1)^2 - 1 = 0$$

**YES**

WHY

Back to 0: it cycles.

Never exceeds 2.

Perform the indicated operation. Write each answer in rectangular form. (3 points each)

**13.**  $\frac{8(\cos 40^\circ + i \sin 40^\circ)}{4(\cos 250^\circ + i \sin 250^\circ)}$

WORK

$$2 \operatorname{cis}(-210^\circ) = 2 \operatorname{cis} 150^\circ$$

$$2 \left( -\frac{\sqrt{3}}{2} + \frac{1}{2}i \right) = \boxed{-\sqrt{3} + i}$$

WHY

Subtract, add 360°.

Perform the indicated operation. Write each answer in rectangular form. (3 points each)

**14.**  $\left[ 4 \operatorname{cis} \frac{5\pi}{3} \right] \left[ 3 \operatorname{cis} \frac{5\pi}{6} \right]$

WORK

$$12 \operatorname{cis} \left( \frac{10\pi}{6} + \frac{5\pi}{6} \right) = 12 \operatorname{cis} \frac{5\pi}{2}$$

$$\frac{5\pi}{2} - 2\pi = \frac{\pi}{2}$$

$$12(0 + i) = \boxed{0 + 12i}$$

WHY

Using  $z$ ,  $z^2 + z$ ,  $(z^2 + z)^2 + z$ , ... is the following number in the Mandelbrot set? Show ALL work. (3 points)

**15.**  $z = -2$

YES / NO

WORK

$$|-2| = 2$$

$$(-2)^2 + (-2) = 2, 2^2 + (-2) = 2, \dots$$

**YES**

WHY

Start.

Stuck at 2.

2 never exceeds 2.

*Trap: Equal to 2 is not exceeding 2.*

Find each power. Write your answer in rectangular form. (3 pts)

**16.**  $(-\sqrt{3} + i)^6$

WORK

$$r = 2, \operatorname{ref} 30^\circ, \text{QII: } 150^\circ$$

$$64 \operatorname{cis} 900^\circ, 900 - 720 = 180^\circ$$

$$64(-1 + 0i) = \boxed{-64 + 0i}$$

WHY

Two full turns.

Find all complex number solutions of each equation. Leave answers in trigonometric form. (4 pts)

**17.**  $x^4 + 16 = 0$

WORK

$$x^4 = -16 = 16 \operatorname{cis} 180^\circ$$

$$2, \frac{180^\circ + 360^\circ k}{4} = 45^\circ + 90^\circ k$$

$$\boxed{2 \operatorname{cis} 45^\circ, 2 \operatorname{cis} 135^\circ, 2 \operatorname{cis} 225^\circ, 2 \operatorname{cis} 315^\circ}$$

WHY

Move 16 over: it becomes negative.

$$k = 0, 1, 2, 3.$$

Convert the polar coordinates to rectangular coordinates. Write your answer in exact form. (3 points)

**18.**  $(-6, \frac{5\pi}{6})$

WORK

$$x = -6 \left( -\frac{\sqrt{3}}{2} \right) = 3\sqrt{3}, \quad y = -6 \left( \frac{1}{2} \right) = -3$$

$$\boxed{(3\sqrt{3}, -3)}$$

WHY

Keep  $r$ 's sign.

QIV.

Convert the rectangular coordinates to polar coordinates with  $r > 0$  and  $0^\circ \leq \theta < 360^\circ$ . (3 points)

**19.**  $(-4, -4)$

WORK

$$r = \sqrt{32} = 4\sqrt{2}$$

$$\text{ref } 45^\circ, \text{ QIII: } 225^\circ$$

$$\boxed{(4\sqrt{2}, 225^\circ)}$$

WHY

*Trap:*  $\tan^{-1}(1) = 45^\circ$  is QI; the point is in QIII.

Convert the rectangular equation to a polar equation. Solve for  $r$ . (3 points)

**20.**  $3x - 4y = 12$

WORK

$$3r \cos \theta - 4r \sin \theta = 12$$

$$\boxed{r = \frac{12}{3 \cos \theta - 4 \sin \theta}}$$

WHY

Find the polar equation of the line that passes through the two points. Simplify your answer. (3 points)

**21.**  $(5, 0^\circ)$  and  $(-5, 90^\circ)$

WORK

$$(5, 0) \text{ and } (0, -5)$$

$$m = \frac{-5 - 0}{0 - 5} = 1, \quad y = x - 5$$

$$x - y = 5$$

$$\boxed{r = \frac{5}{\cos \theta - \sin \theta}}$$

WHY

Negative  $r$  at  $90^\circ$  lands below.

Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

**22.**  $r = -6 \cos \theta$

WORK

$$x^2 + y^2 = -6x$$

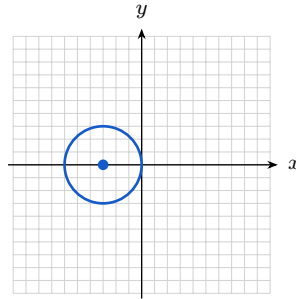
$$x^2 + 6x + 9 + y^2 = 9$$

$$(x + 3)^2 + y^2 = 9$$

WHY

Multiply by  $r$ , substitute.

Center  $(-3, 0)$ , radius 3.



Convert each polar equation to a rectangular equation and graph it. (No calculator) (4 points each)

**23.**  $r = \frac{8}{\cos \theta - 2 \sin \theta}$

WORK

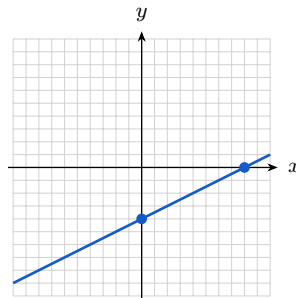
$$x - 2y = 8$$

$$y = \frac{1}{2}x - 4$$

WHY

Clear fraction, substitute.

Intercepts  $(8, 0)$ ,  $(0, -4)$ .



Find the polar coordinates of the points of intersection for the given interval of  $\theta$ . (4 points)

**24.**  $r = 3 + 2 \sin \theta$ ,  $r = 3 + 2 \cos \theta$   
 $0 \leq \theta < 2\pi$

WORK

$$\sin \theta = \cos \theta: \theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

$$\frac{\pi}{4}: r = 3 + \sqrt{2}; \frac{5\pi}{4}: r = 3 - \sqrt{2}$$

$$\left(3 + \sqrt{2}, \frac{\pi}{4}\right), \left(3 - \sqrt{2}, \frac{5\pi}{4}\right)$$

WHY

Set  $r = r$ ; the 3s cancel.

Plug back in.

$r \geq 1$ : no pole.

Identify the type of polar graph. Use the word bank: circle, cardioid, limaçon, rose curve, lemniscate. (2 points)

**25.**  $r = 4 + 3 \sin \theta$

WORK

limaçon

WHY

$$\frac{a}{b} = \frac{4}{3} \neq 1.$$